Indian Institute of Technology, Kanpur

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Laboratory Report

Antenna Characterization

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Degree programme:	PhD
Submission date:	October 6, 2026

1 Objective

(a) To determine the antenna operating frequency and return loss from VSWR measurements.

(b) To measure the effect of distance on the power received by the antenna.

(c) To measure the far-field power pattern of the given antenna with respect to angle in both co-polarization and cross-polarization configurations.

(d) To determine the half-power beamwidth and estimate the directivity of the antenna.

2 Theory

2.1 Basic antenna concepts

An antenna is a transducer that converts electrical power into electromagnetic waves and vice versa. A dipole antenna has two equal metal arms in a straight line and is fed at the centre. Its total length is approximately half a wavelength ($\lambda/2$). A monopole antenna is approximately half of a dipole mounted above a conducting ground plane, making it behave like a full dipole with a length of approximately one-quarter wavelength ($\lambda/4$).

2.2 Antenna beam pattern

The energy radiated by an antenna is represented by the radiation pattern, which generally features a main lobe, side lobes, and a back lobe. The main lobe is the region where the maximum radiated energy exists. The other parts of the pattern where radiation is distributed sideways are known as side lobes, while the lobe opposite to the main lobe is known as the back lobe. E-plane and H-plane angular measurements are used to determine this pattern. The E-plane is the plane containing the electric-field vector, and the H-plane is the plane containing the magnetic-field vector.

2.3 Reflection coefficient, VSWR, and return loss

When the antenna impedance Z_L differs from the characteristic impedance Z_0 of the transmission line, a part of the incident signal is reflected. The voltage reflection coefficient at the antenna terminals is:

$$\Gamma = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

The magnitude $|\Gamma|$ indicates the degree of impedance mismatch. The voltage standing-wave ratio is related to the magnitude of the reflection coefficient by:

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

The return loss is defined as:

$$\text{RL} = -20 \log_{10} |\Gamma| \text{ dB}$$

A well-matched antenna has a VSWR close to unity, a small reflection coefficient, and a large return-loss value.

2.4 Half-power beamwidth and directivity

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two points where the received power drops to half of its maximum value, corresponding to the -3 dB points. Directivity is a measure of how concentrated the radiation is in the direction of the main lobe compared with an isotropic antenna. The directivity is related to the beam solid angle Ω_A as:

$$D = \frac{4\pi}{\Omega_A}$$

For practical antennas, it can be approximated using the half-power beamwidths in the E-plane (θ_E) and H-plane (θ_H) expressed in radians as:

$$D \approx \frac{4\pi}{\theta_E \theta_H}$$

3 Components Required

Monopole and dipole antennas, RF signal generator, digital storage oscilloscope (DSO), spectrum analyzer, RF Tee connector, 50 Ω termination, antenna support stand, coaxial cables, and suitable RF adapters.

4 Experimental Setup and Procedure

- **VSWR and operating frequency:** The signal generator, Tee connector, antenna, and oscilloscope were connected to record V_{max} and V_{min} across a frequency sweep to calculate VSWR and identify the resonant frequency.
- **Effect of distance:** With aligned antennas at a fixed height, the distance was varied and the received power was recorded.
- **Co-polarization beam pattern:** The E-plane and H-plane angles were varied from -90° to $+90^\circ$ to record received power in a coplanar alignment.
- **Cross-polarization beam pattern:** The transmitting antenna was adjusted so the electric fields were perpendicular, and the angular measurements were repeated.
- **Half-power beamwidth and directivity:** The -3 dB points were identified from the co-polarized plots to calculate θ_E and θ_H, allowing for the estimation of directivity.

5 Observations and Results

5.1 Antenna Operating Frequency and Far-Field Boundary

Based on the physical dimensions of the antenna, the theoretical operating frequency was calculated. For a quarter-wave ($\lambda/4$) monopole antenna with a measured length of 5 cm (0.05 m):

$$\frac{\lambda}{4} = 0.05 \text{ m} \implies \lambda = 0.2 \text{ m}$$

$$f_{\text{operating}} = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{0.2 \text{ m}} = 1.5 \text{ GHz}$$

To ensure proper beam pattern measurements, the minimum far-field distance (Fraunhofer region) was verified using the maximum aperture dimension ($D = 0.29$ m) and the signal generator power set to 20 dBm:

$$R_{\text{far-field}} > \frac{2D^2}{\lambda} = \frac{2(0.29)^2}{0.2} = 0.841 \text{ m}$$

All angular measurements were taken beyond this 84.1 cm boundary.

5.2 VSWR and Return Loss

For this portion of the experiment, an adjustable antenna was utilized and tuned to operate in the 200 MHz region. Instead of using fixed frequency intervals, a dynamic frequency sweep was performed to specifically identify and record the frequencies where local voltage maxima (V_{max}) and minima (V_{min}) occurred on the oscilloscope. The Voltage Standing Wave Ratio (VSWR), Reflection Coefficient magnitude ($|\Gamma|$), and Return Loss (RL) were calculated using these targeted extreme values through the following relations:

$$\text{VSWR} = \frac{V_{max}}{V_{min}}$$

$$|\Gamma| = \frac{\text{VSWR} - 1}{\text{VSWR} + 1} = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$$

$$\text{RL} = -20 \log_{10} |\Gamma| \text{ dB}$$

Frequency (MHz)	V_{max} (mV)	V_{min} (mV)	VSWR	Return Loss (dB)
40	485	340	1.43	15.10
60	485	377	1.29	18.04
80	456	377	1.21	20.46
110	456	246	1.85	10.48
130	327	246	1.33	16.99
150	327	240	1.36	16.28
170	304	240	1.27	18.59
190	304	190	1.60	12.74
220	207	190	1.09	27.37
240	207	161	1.29	18.06
270	187	161	1.16	22.53
300	187	133	1.41	15.45

Table 1: *VSWR and Return Loss measurements across frequency*

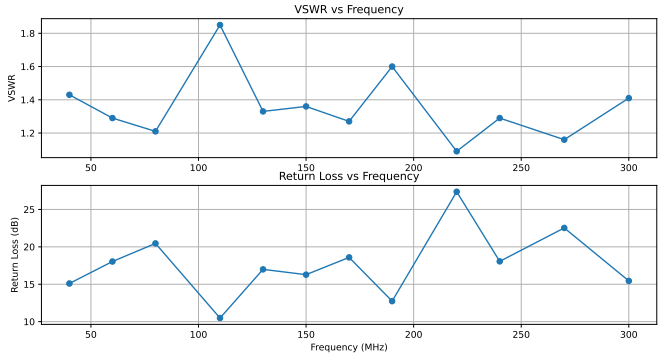


Figure 1: *VSWR and Return Loss versus Frequency*

The minimum VSWR observed is 1.09, which corresponds to a maximum return loss of 27.37 dB. This optimal matching occurs at an operating frequency of 220 MHz. The VSWR successfully remained below 2.0 for the entire tested frequency range between 40 MHz and 300 MHz.

5.3 Effect of distance

The received power was recorded at various distances to observe path loss. The orientation of antenna was in co-polarization with E-plane.

Distance (R) [cm]	$20\log_{10}(R)$	**Received Power (P_r) [dBm]**
118	41.44	-19.38
221	46.89	-26.38
318	50.05	-27.80
416	52.38	-28.27
515	54.24	-29.91
615	55.78	-34.08
715	57.09	-39.03

Table 2: *Received power at varying distances between transmitting and receiving antennas*

The received power was plotted against $20\log_{10}(R)$. The resulting linear relationship demonstrates that the path loss of the signal mathematically follows the inverse-square law as the distance between the antennas increases.

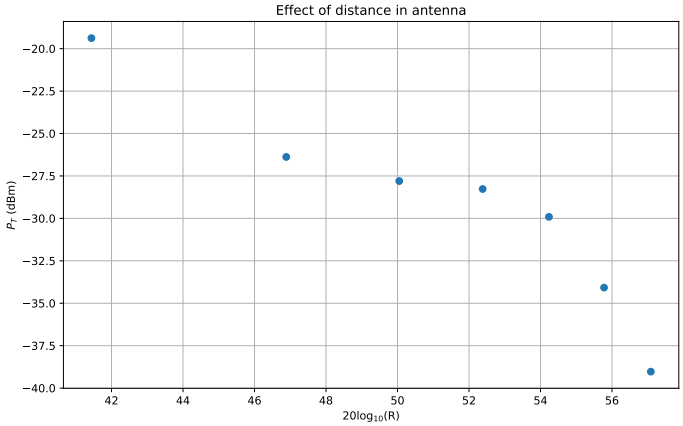


Figure 2: *Received power vs* $20\log_{10}(R)$

5.4 Beam pattern measurements

The received power was measured at 20° intervals for both the E-plane and H-plane in co-polarization and cross-polarization configurations.

Angle (°)	Co-polarization (dBm)		Cross-polarization (dBm)	
	E-plane	H-plane	E-plane	H-plane
0	-56.87	-28.50	-51.75	-28.52
20	-49.59	-30.88	-44.50	-30.50
40	-47.55	-37.39	-41.00	-34.20
60	-54.38	-44.46	-43.51	-36.70
80	-52.82	-46.17	-49.45	-42.30
100	-55.44	-52.92	-49.55	-45.05
120	-54.42	-45.48	-47.60	-47.12
140	-51.49	-45.45	-53.95	-45.75
160	-52.19	-43.24	-53.15	-42.30
180	-54.65	-37.49	-52.15	-45.35
200	-49.76	-39.55	-46.56	-45.15
220	-51.39	-44.88	-52.50	-51.40
240	-50.30	-45.70	-47.60	-48.07
260	-51.20	-40.47	-54.00	-48.80
280	-46.87	-41.17	-48.90	-43.40
300	-58.21	-35.32	-45.70	-38.80
320	-54.67	-31.80	-48.41	-33.35
340	-51.93	-29.90	-50.60	-34.70
360	-56.87	-28.50	-51.75	-28.52

Table 3: *Angular received power measurements for antenna beam patterns*

The radiation patterns were visualized using polar projections to compare the spatial distribution of the main and minor lobes.

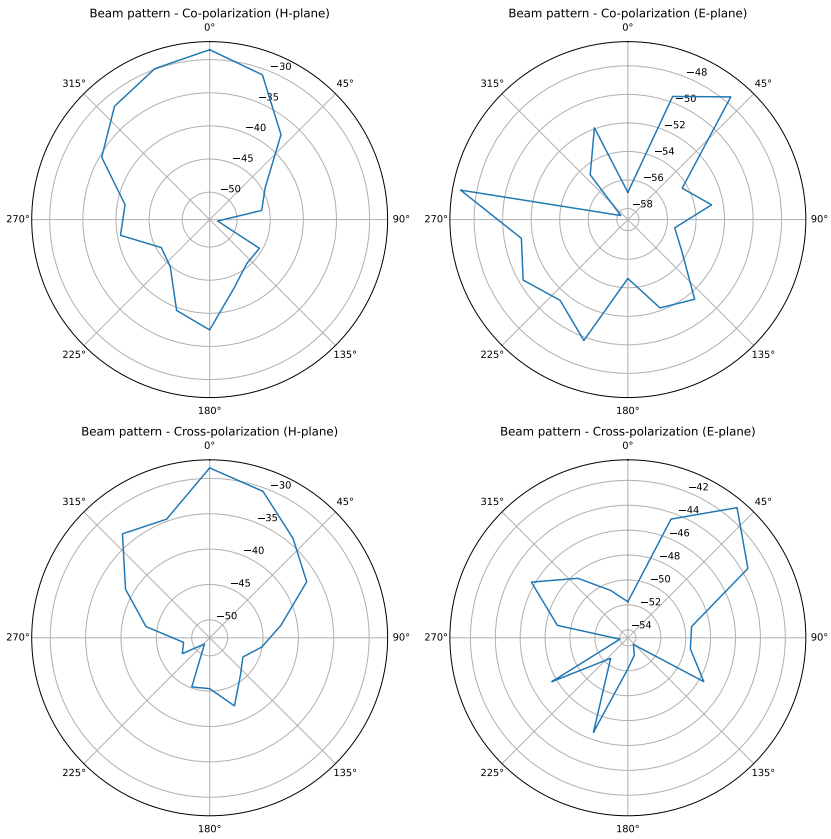


Figure 3: *Antenna beam patterns in polar coordinates*

5.5 Half-power beamwidth and directivity

The −3 dB threshold used to determine the angular width of the main lobes is visualized as a red dashed concentric line on the co-polarization beam patterns.

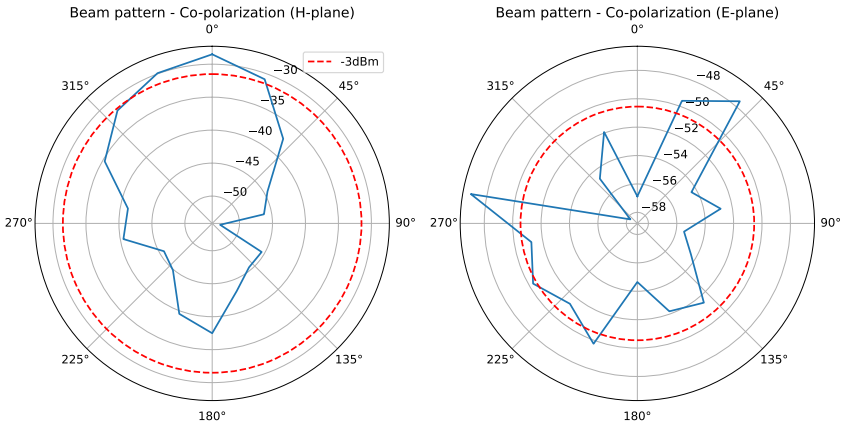


Figure 4: *Co-polarization beam patterns showing the -3 dB half-power threshold*

Linear interpolation of the co-polarization arrays yielded the following half-power beamwidths:

- **E-plane HPBW (θ_E):** 31.4° (0.548 rad). (Note: The main lobe in the E-plane exhibited a 45° mechanical offset relative to the true electrical peak during the experimental setup).

- **H-plane HPBW (θ_H):** 58.7° (1.024 rad).
- **Directivity (D):** $D \approx \frac{4\pi}{\theta_E \theta_H} = \frac{4\pi}{0.548 \times 1.024} = 22.4$.

Converting the absolute directivity ratio to decibels gives an estimated directivity of approximately **13.5 dB**.

6 Results and Conclusion

The antenna characterization experiment successfully evaluated the radiation properties and path loss of the provided antenna. The key findings are summarized below:

- **Operating Frequency and VSWR (Part A):** Using an antenna tuned for the VHF band, the VSWR and return loss were successfully measured across a 40 MHz to 300 MHz sweep. The antenna exhibited a minimum VSWR of 1.09 and a peak return loss of 27.37 dB, identifying the experimental operating frequency as around 220 MHz. The VSWR remained below 2.0 throughout the entire tested spectrum, indicating a broadband impedance match.
- **Effect of Distance (Part B):** The received power decayed logarithmically with distance. Plotting the power against $20 \log_{10}(R)$ produced a highly linear negative slope, experimentally verifying that the antenna's far-field path loss strictly follows the inverse-square law.
- **Beam Patterns and Polarization (Parts C & D):** The co-polarization measurements revealed distinct main radiation lobes in both the E-plane and H-plane. Conversely, the cross-polarization configuration resulted in significantly lower received power across all angles and lacked a defined main beam. This stark contrast demonstrates the antenna's strong polarization selectivity.
- **HPBW and Directivity (Part E):** Using linear interpolation, the half-power beamwidths were determined to be $\theta_E = 31.4°$ and $\theta_H = 58.7°$. This yielded an estimated directivity of $D \approx 22.4$, or 13.5 dB, indicating a highly directional radiated beam.
- **Measurement Uncertainty & Data Representation:** The E-plane co-polarization plot exhibited a 45° main lobe shift, which was an artifact of mechanical misalignment during the physical setup. The jagged secondary lobes in the polar plots are primarily attributed to indoor multipath interference (RF reflections). Furthermore, the angular, oscillating appearance of the VSWR plot is a direct result of the measurement methodology: because data points were specifically recorded at the frequencies corresponding to local voltage extrema (peaks and valleys), the resulting plot connects these turning points, illustrating the envelope of the impedance ripples rather than a continuous, smooth resonance curve.